**INTERNATIONAL QUANTITATIVE INVESTING CHALLENGE 2026**

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# Strategy Declaration

ML Supertrend: Adaptive Reversal Strategy


**Young Investors Cup 2026 gold medallists**

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## Investment hypothesis and complete strategy

The strategy seeks short-term rebounds after unusually strong selling in volatile stocks. A fresh low alone is not a BUY: price must confirm an upward Supertrend reversal, recent oversold RSI and signal spacing. The machine-learning layer learns online from completed hypothetical long and short probes. Three model components use rolling outcomes, adverse-movement-weighted batches and an 8 by 8 market-context memory to adjust the Supertrend multiplier and ATR smoothing within bounds. The learning changes signal sensitivity and timing; explicit portfolio rules still set entry, position size and exit. The hypothesis fails when selling reflects lasting information, or costs erase any rebound.

Each probe opens at a candle open and matures at the close of its fifth candle. A move of one entry ATR gives the long probe a score of +1 and the matching short probe -1. Only matured outcomes can update later settings. Because probes overlap, their labels are dependent and are not account returns.

At equal confidence, one ATR of adverse movement halves a probe's batch weight. Context-memory influence rises with its supporting observations but is capped at 65%. A bounded update layer smooths parameter proposals; portfolio rules separately set entries, size and exits.

## Why these stocks

DELL is the paper-trading example. The strategy scans liquid S&P 500 stocks; each BUY must pass liquidity and portfolio risk checks before an order is sized.

# 1 Strategy and quantitative method

Use completed regular-session 30-minute candles for supported S&P 500 stocks. BUY opens one long position; repeated BUYs do not add size. SELL or an earlier protective stop closes it. Orders follow confirmation; historical replay fills at the next bar open plus costs.

## 1. Measure normal price movement

**True Range Equation**

$$TR_t = \max\{H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\}$$

t is the current completed candle; t - 1 is the previous candle. H, L and C are high, low and close in dollars. TR takes the largest candle range or gap from the previous close. Vertical bars mean absolute value.

**Volatility Envelope Equations**

$$x_t = \frac{H_t + L_t + 2C_t}{4}, \qquad A_t = A_{t-1} + \frac{TR_t - A_{t-1}}{\ell_t}$$
$$b_t^- = x_t - m_t A_t, \qquad b_t^+ = x_t + m_t A_t$$

x is the close-weighted price, A the smoothed true range, l the smoothing length and m the multiplier. The boundaries sit mA above and below x; the baseline uses l = 24 and m = 1.4. They trail or reset around price. A close across the previous opposing band flips trend.

## 2. Machine-learning models and completed probes

**Probe Return Equation**

$$g = s\frac{C - P}{A_{\text{entry}}}$$

g is the probe score; s is +1 for a long and -1 for a short. P is entry price, C the fifth-candle close and A entry the smoothed true range at entry. A one-ATR price rise gives g = +1 for the long and -1 for the short. Up to five probes per direction can overlap; each result enters learning after its fifth candle closes.

Three online model components use matured probes. The rolling-performance learner tracks a 30-outcome baseline window per direction: below 38% wins with negative mean score, it proposes wider bands and slower ATR smoothing; above 62% with a positive mean, tighter and faster settings. The micro-batch learner weighs each result by confidence divided by one plus adverse excursion in ATR units, then assesses normalized gain, profit factor and downside ratio. The context-memory learner maps trend/range state and relative volatility to an 8 by 8 grid. Hurst persistence, inverse entropy and ADX contribute to the regime score; smoothed cell proposals blend with Gaussian-distance-weighted neighbors. Its influence grows with sample count, capped at 0.65.

A meta-combiner blends rolling and context proposals, then incorporates available batch recommendations. A larger multiplier requires a larger reversal to flip trend; a longer ATR length reacts more slowly to volatility. The model uses deadbands, update caps, bounded parameter ranges and periodic reversion to its baseline. Portfolio rules set account exposure.

**Adaptive Update Equation**

$$\theta_{t+1} = \theta_t + \alpha_t(\tilde{\theta}_t - \theta_t)$$

theta is the current multiplier or ATR length; theta tilde is the rounded, bounded target; alpha is the update fraction. For multiplier 1.40, target 1.50 and alpha 0.20, the update is 1.42. Deadbands ignore tiny changes; caps constrain each update and periodic reversion pulls settings toward baseline. The multiplier stays 0.5-5.0 and ATR length 5-100.

## 3. Turn the adjusted indicator into a decision

BUY requires a stored fresh low, an upward trend flip, RSI(14) below 30 within 50 candles and at least 10 candles since the last signal. RSI compares smoothed gains and losses on a 0-100 scale. SELL uses a fresh-high reversal, downward flip and recent RSI above 70. Volume-surge filtering and live tick pressure are off; all decisions use confirmed candles.

**Learning check.** Across 780 DELL bars, 776 probes per side matured. The multiplier moved from 1.40 to 2.145 while ATR length remained 24; the model produced 6 BUY and 8 SELL signals. Prefix checks matched. This verifies the calculation sequence, not predictive performance. Data: Yahoo Finance.

# 2 ML Supertrend replay and backtest charts

We replayed ML Supertrend BUY emails from 8 September 2025 to 14 October 2026, carrying the last list through 30 October (290 sessions). Each list was applied at the next session open, starting with $100,000 and charging 10 basis points per side. Historical sizing allowed up to ten holdings; the current declaration caps the portfolio at five.

| ML Supertrend replay result | Value |
|---|---|
| Ending marked equity | $151,542.68 |
| Total return / annualized growth | +51.54% / +43.92% |
| Maximum drawdown | -21.10% |
| QQQ benchmark return | +19.39% |

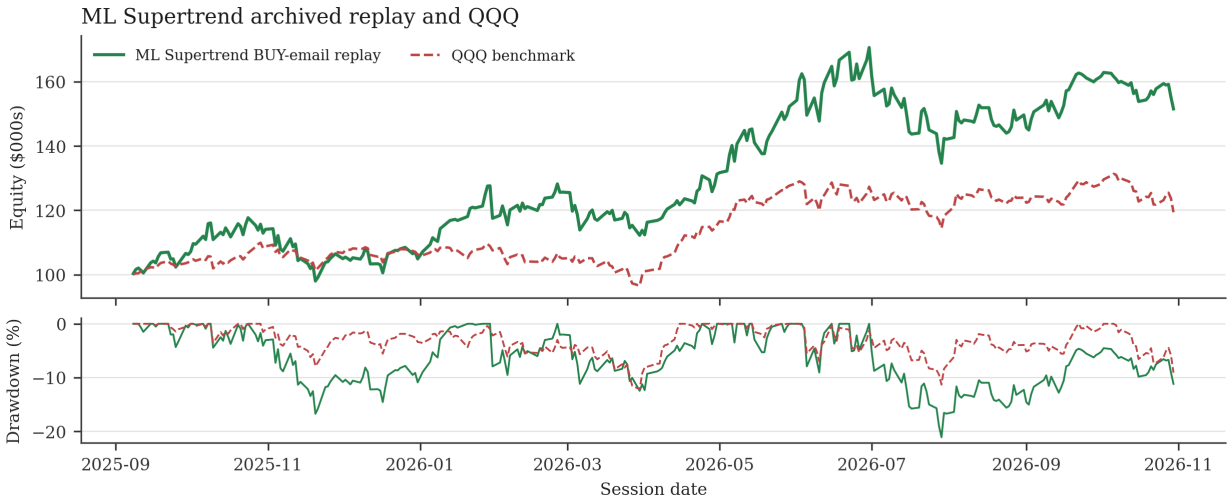



*Figure 1. ML Supertrend archived replay and QQQ benchmark: marked equity and peak-to-trough drawdown. Data: Alpaca SIP split-adjusted daily bars; our replay calculations.*

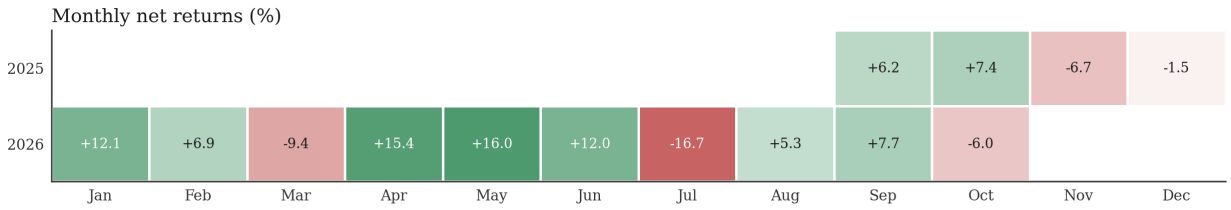



*Figure 2. Monthly net returns (%); September 2025 is a partial month. Data: same replay.*

Sensitivity: return was +39.11% at 20 bps per side and +43.49% with a one-session ranking delay. All 1,254 saved fills and 290 daily marks reproduce. Rules and source hashes: archived backtest documentation.

# 3 Trade expectancy and risk tests

**Net Trade Expectancy Equation**

$$e = pW - (1-p)D, \qquad p_{\mathrm{BE}} = \frac{D}{W+D}$$

e is mean net return per closed trade; p is the net win rate; W is the average net winning return and D the positive magnitude of the average net losing return. All returns use percentage units and already include costs. The break-even win rate, p BE, is the win rate needed to offset losses at those payoff sizes.

| Trade statistic | Historical result |
|---|---|
| Completed trades | 622 |
| Win rate | 46.62% |
| Mean net profit / return per trade | +$81.73 / +0.8173% |
| Median trade return | -0.6268% |
| Average net winning / losing return | +8.05% / -5.50% |
| Profit factor | 1.28 |
| Break-even win rate at observed payoffs | 40.59% |
| Best / worst net trade return | +62.80% / -40.03% |
| Longest losing streak | 15 trades |

These 622 closed trades come from the ML Supertrend email replay. The 46.62% win rate exceeded its 40.59% break-even level; profit factor was 1.28. The median trade lost 0.6268%, while larger gains lifted the mean. Open positions are excluded. Trade calculations and fills: GitHub trade evidence.

## ML Supertrend one-year Monte Carlo

We resampled the daily equity returns from the archived ML Supertrend replay: 290 sessions, 8 September 2025–30 October 2026. Circular five-session blocks preserve short runs of gains and losses. The 50,000 paths each span 252 sessions. The replay source, calculations and chart are documented in the Monte Carlo files.

| One-year path statistic | Result |
|---|---|
| Median terminal return | +43.98% |
| 5th / 95th percentile terminal return | -11.85% / +131.43% |
| Paths ending below start | 10.87% |
| Mean return in worst 5% of paths | -21.93% |
| 95th percentile maximum drawdown | 35.46% |

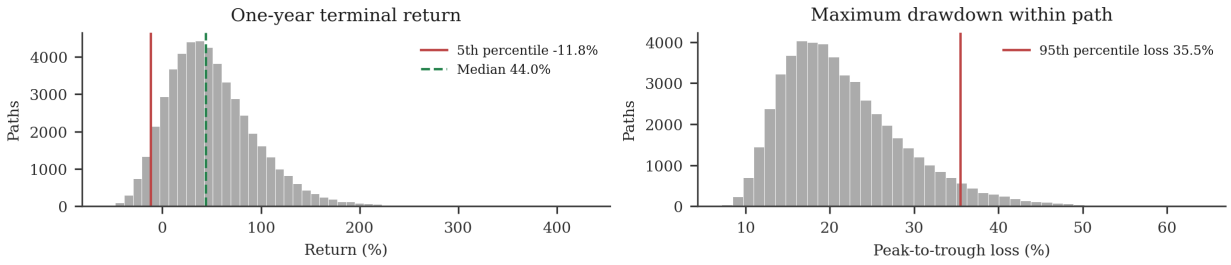



*Figure 3. One-year terminal returns and maximum drawdowns, resampled from the 290-session ML Supertrend replay.*

# 4 Portfolio construction and risk

False reversals can cluster during persistent downtrends; overnight gaps can cross stops, and technology names can fall together. Entry spacing, planned-loss budgets, sector/theme caps and the account halt address these risks. All limits below apply together and are declared design choices, not optimized thresholds.

| Control | Execution rule and purpose |
|---|---|
| Exposure | Long-only, no borrowing; at most five names, 15% of equity per name and 75% gross stocks. Five names at the individual cap use the full portfolio allowance. |
| Loss budget | Initial stop two 30-minute ATR(24) below fill; planned risk at most 0.5% of equity per name and 2% across the account. Never widen a stop. |
| Concentration | At most 20% in one sector and 25% in the combined semiconductor / technology-infrastructure cluster. Correlated names consume shared capacity. |
| Liquidity | Price at least $5; trailing 20-session average dollar volume at least $20 million. Reject stale, halted or incomplete quotes. |
| Account losses | At 2% session loss, pause entries until the next session. At 6% from peak marked equity, liquidate at available prices and halt for the competition. |
| Execution integrity | Use one account; reject duplicate BUY entries. Log accepted/rejected signals, settings, quotes, order and fill times, fees, stop and exit reason. Freeze the declared rules; close by the competition cutoff. |

**Position Sizing Equation**

$$d_i = 2\,\mathrm{ATR}_{24,i}, \qquad q_i = \left\lfloor \min\left(\frac{0.15E}{P_i}, \frac{0.005E}{d_i}\right) \right\rfloor$$

For stock i, E is marked account equity, P its entry price, ATR with subscript 24 its 24-candle smoothed true range, d the dollar stop distance and q whole shares. Floor brackets round down. The smaller value/risk cap governs; remaining exposure, sector capacity, aggregate risk and costs can reduce the order.

At the recorded DELL close of $563.17, a $15,000 cap permits 26 shares ($14,642.42). The $500 loss budget in a $100,000 account allows at most $19.23 stop distance at that size. A wider two-ATR stop reduces shares. The 15% name limit and five-position limit allow at most 75% gross exposure; remaining equity stays in cash. Position size can be lower when the stop-risk budget binds.

## Gap and execution stress

| Stop execution stress | Loss / planned stop loss | Account loss at 2% risk budget |
|---|---|---|
| Fills at intended stop | 1.0 x | 2% |
| Loss exceeds stop distance by 50% | 1.5 x | 3% |
| Loss doubles the intended distance | 2.0 x | 4% |
| Gap produces three times planned loss | 3.0 x | 6% |

The table multiplies planned account risk by the realized stop-distance overrun. A gap can exceed every scenario shown, so the 6% halt trigger cannot cap losses.

Stress grid: 96 combinations of horizon, exposure, volatility and correlation (100,000 paths each); daily Gaussian, t(5), gap and liquidation tests use 40,000 paths per model. At the 75% ceiling, common 10%/20% price declines produce 7.5%/15% account losses before costs. Opening gaps can exceed the 6% halt threshold.

## Final declared strategy

Trade confirmed 30-minute ML Supertrend reversals, long-only. Matured probes update bounded settings; qualifying BUYs pass the sizing and portfolio limits above. Hold at most five names, 15% per name and 75% total stocks. Exit on SELL or stop; pause entries at 2% session loss and liquidate and halt at 6% account drawdown.